VivaGuard: Adaptive AI Oral Defense and Authentic Comprehension Platform

Formal Abstract & Comprehensive Literature Survey

***Abstract*— *Modern generative AI models (LLMs) present a profound threat to academic evaluation integrity. Existing statistical AI detectors suffer from severe false-positive rates—disproportionately penalizing non-native English writers—and remain vulnerable to adversarial paraphrasing. This paper introduces **VivaGuard**, an evidence-based academic integrity platform that shifts the paradigm from punitive text scanning to authentic document-grounded oral defense (viva voce). VivaGuard extracts a structured Submission Knowledge Map from student papers (PDF, DOCX, TXT) and orchestrates a 3-to-10 minute adaptive voice examination. The system evaluates responses across six cognitive dimensions: Conceptual Understanding (30%), Methodology (20%), Reasoning (15%), Critical Thinking (15%), Submission Familiarity (15%), and Communication (5%). Rather than issuing automated cheating accusations, VivaGuard provides faculty with grounded evidence reports featuring exact page/paragraph citations and audio transcripts.*

1. Literature Survey & Related Work

The literature survey evaluates prior research across four key pillars: the statistical failures of text-based AI detectors, the pedagogical foundations of oral viva assessment, retrieval-augmented generation (RAG) for hallucination prevention, and adaptive educational state engines.

1.1 Limitations of Statistical AI Detection Models

Recent empirical literature demonstrates the fundamental unreliability of statistical classifiers for detecting AI-generated text:

- **High False-Positive Bias on Non-Native Writers:** Liang et al. (2023) evaluated seven commercial AI detectors on TOEFL essays written by non-native English speakers. The detectors misclassified over 61% of authentic human essays as AI-generated due to lower linguistic perplexity and structural uniformity.
- **Adversarial Evasion Vulnerability:** Sadasivan et al. (2023) theoretically proved that as LLM text distributions approach human capability, statistical detection accuracy approaches random chance. Light paraphrasing tools reduce detection sensitivity to near zero.
- **Pedagogical Incompatibility:** Dawson & Sutherland-Smith (2018) highlighted that arbitrary probability scores damage student-faculty trust without providing diagnostic learning evidence.

1.2 Pedagogical Foundations of Oral Examinations (Viva Voce)

Oral defenses represent the gold standard of authentic academic assessment:

- **Bloom's Taxonomy Higher-Order Evaluation:** Bloom et al. (1956) and Anderson & Krathwohl (2001) establish that while written papers can be ghostwritten or generated, live oral defenses force candidates to operate at higher cognitive levels: Analysis, Evaluation, and Defense.
- **Combatting Ghostwriting & Contract Cheating:** Joughin (1998, 2010) and Brown & Knight (1994) demonstrated that oral vivas effectively eliminate contract cheating because third-party proxy writers cannot replicate personal familiarity under spontaneous questioning.

• **The Scalability Bottleneck:** Traditionally, oral vivas were restricted to doctoral dissertations due to high faculty time demands. VivaGuard automates viva orchestration while maintaining human faculty oversight.

1.3 Comparative System Matrix

Dimension	Statistical Detectors (Turnitin/GPTZero)	Human Viva Voce	VivaGuard Platform
Core Paradigm	Statistical text scanning	Live faculty dialogue	**Document-Grounded AI Viva**
Primary Output	Accusatory probability %	Qualitative pass/fail	**0-100 Score + Evidence Report**
Grounding	Text n-gram matching	Human memory of paper	**Knowledge Map (Page & Para Ref)**
Scalability	High (Instant scan)	Low (Faculty hours)	**High (Automated 100+ vivas)**
Non-Native Bias	Severe (>61% false positive)	Low	**Low (Tests understanding)**

2. VivaGuard Methodological Contributions

1. **Knowledge Map Tree Extraction:** Automated extraction of claims, methodologies, tools, and citations into an inspectable tree graph.
2. **Document-Grounded Question Pool:** Categorized candidate questions spanning 8 question types (Understanding, Methodology, Defense, Application, Critical Thinking, Citation, Deep Dive, Contradiction).
3. **Adaptive Examination Engine:** Real-time Speech-to-Text (STT) and Text-to-Speech (TTS) orchestration with dynamic follow-up questioning.
4. **Human-in-the-Loop Governance:** Faculty dashboard with exact source page/paragraph references, audio transcripts, and score override capabilities.

3. References

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